

Lista V

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Inicialização

```
library(DBI)
library(RSQLite)

CON <- dbConnect(RSQLite::SQLite(), "../datasets/me315.sqlite")
```

Respostas

Sala de aula

1.

```
SELECT FEE, FEE + 1.5, FEE + 1, FEE * 2, FEE / 2.0
FROM PRESERVE
WHERE STATE = 'AZ'
```

Table 1: 4 records

FEE	FEE + 1.5	FEE + 1	FEE * 2	FEE / 2.0
3	4.5	4	6	1.5
0	1.5	1	0	0.0
3	4.5	4	6	1.5
3	4.5	4	6	1.5

2.

```
SELECT PNAME, ACRES, ACRES / 2.0, ACRES / 2
FROM PRESERVE
WHERE STATE = 'MT'
```

Table 2: 4 records

PNAME	ACRES	ACRES / 2.0	ACRES / 2
DANCING PRAIRIE	680	340.0	340
SOUTH FORK MADISON	121	60.5	60
COMERTOWN PRAIRIE	1130	565.0	565
PINE BUTTE SWAMP	15000	7500.0	7500

Divisão por inteiro retorna apenas a parte inteira da divisão, não o valor exato como a divisão por float.

Hands-On

1.

```
SELECT PNAME, ACRES, ACRES * 1.2
FROM PRESERVE
```

Table 3: Displaying records 1 - 10

PNAME	ACRES	ACRES * 1.2
HASSAYAMPA RIVER	660	792.0
DANCING PRAIRIE	680	816.0
MULESHOE RANCH	49120	58944.0
SOUTH FORK MADISON	121	145.2
MCELWAIN-OLSEN	66	79.2
TATKON	40	48.0
DAVID H. SMITH	830	996.0
MIACOMET MOORS	4	4.8
MOUNT PLANTAIN	730	876.0
COMERTOWN PRAIRIE	1130	1356.0

2.

Como comentário, as colunas definidas por AS, as aliases, podem ser utilizadas em ORDER_BY porém não em WHERE.

```
SELECT PNAME, ACRES, ACRES / 3.0 AS ADJUSTED_ACRES
FROM PRESERVE
```

Table 4: Displaying records 1 - 10

PNAME	ACRES	ADJUSTED_ACRES
HASSAYAMPA RIVER	660	220.000000
DANCING PRAIRIE	680	226.666667
MULESHOE RANCH	49120	16373.333333
SOUTH FORK MADISON	121	40.333333
MCELWAIN-OLSEN	66	22.000000
TATKON	40	13.333333
DAVID H. SMITH	830	276.666667
MIACOMET MOORS	4	1.333333
MOUNT PLANTAIN	730	243.333333
COMERTOWN PRAIRIE	1130	376.666667

3.

```
SELECT PNAME, FEE, (FEE + 50) / 2.0 AS ADJUSTED_FEE
FROM PRESERVE
```

Table 5: Displaying records 1 - 10

PNAME	FEE	ADJUSTED_FEE
HASSAYAMPA RIVER	3	26.5
DANCING PRAIRIE	0	25.0
MULESHOE RANCH	0	25.0
SOUTH FORK MADISON	0	25.0
MCELWAIN-OLSEN	0	25.0
TATKON	0	25.0
DAVID H. SMITH	0	25.0
MIACOMET MOORS	0	25.0
MOUNT PLANTAIN	0	25.0

PNAME	FEE	ADJUSTED_FEE
COMERTOWN PRAIRIE	0	25.0

4.

```
SELECT AVG(ACRES) AS MEAN_ACRES, MIN(ACRES) AS MIN_ACRES, MAX(ACRES) AS MAX_ACRES
FROM PRESERVE
WHERE STATE = 'AZ'
```

Table 6: 1 records

MEAN_ACRES	MIN_ACRES	MAX_ACRES
12840	380	49120

5.

```
SELECT MIN(PNAME) AS FIRST_NAME
FROM PRESERVE
-- ORDER BY PNAME
-- LIMIT 1
```

Table 7: 1 records

FIRST_NAME
COMERTOWN PRAIRIE

6.

```
SELECT COUNT(DISTINCT FEE) AS DISTINCT_FEES
FROM PRESERVE
WHERE FEE != 0
```

Table 8: 1 records

DISTINCT_FEES
1

7.

```
SELECT COUNT(PNAME) AS NAMES, COUNT(DISTINCT PNAME) AS DISTINCT_NAMES
FROM PRESERVE
```

Table 9: 1 records

NAMES	DISTINCT_NAMES
14	14

Todos nomes são distintos, uma vez que os dois conjuntos tem mesma contagem e nomes distintos são subconjunto de nomes.

8.

```
SELECT AVG(ACRES * 0.02) AS MEAN_NEW_FEE
FROM PRESERVE
WHERE STATE = 'AZ'
```

Table 10: 1 records

MEAN_NEW_FEE
256.8

9.

```
SELECT ENO, ENAME, SALARY AS OLDSALARY, SALARY * 1.10 AS NEWSALARY
FROM EMPLOYEE
ORDER BY NEWSALARY
```

Table 11: 6 records

ENO	ENAME	OLDSALARY	NEWSALARY
5000	JOE	400	440
4000	SHEMP	500	550
1000	MOE	2000	2200
2000	LARRY	2000	2200
3000	CURLY	3000	3300
6000	GEORGE	9000	9900

10.

```
SELECT ENO, ENAME, SALARY AS OLDSALARY, SALARY * 1.10 AS NEWSALARY
FROM EMPLOYEE
WHERE SALARY * 1.10 > 2000
ORDER BY NEWSALARY
```

Table 12: 4 records

ENO	ENAME	OLDSALARY	NEWSALARY
1000	MOE	2000	2200
2000	LARRY	2000	2200
3000	CURLY	3000	3300
6000	GEORGE	9000	9900

11.

```
SELECT SUM(SALARY), AVG(SALARY), MAX(SALARY), MIN(SALARY)
FROM EMPLOYEE
```

Table 13: 1 records

SUM(SALARY)	AVG(SALARY)	MAX(SALARY)	MIN(SALARY)
16900	2816.667	9000	400

12.

```
SELECT COUNT(*)  
FROM EMPLOYEE  
WHERE DNO == 20
```

Table 14: 1 records

COUNT(*)
3

13.

```
SELECT COUNT(DISTINCT DNO)  
FROM EMPLOYEE
```

Table 15: 1 records

COUNT(DISTINCT DNO)
3